Jonathan Lee

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Education

Northwestern University — Evanston, IL
B.S. in Computer Science and Mathematics (Pure Mathematics concentration) — *Expected June 2027*
- GPA: 3.84
- CS Honors Track and Research Track
- McCormick Dean's List (5 of 6 quarters)

University of Kansas — Lawrence, KS
Undergraduate coursework, transferred to Northwestern — *August 2023 – May 2024*
- GPA: 4.00
- Albert F. and Barbara Kile Zernickow Mathematics Scholarship

Research Experience

Researcher, Cryptography — March 2026 – Present
Advisor: Prof. Xiao Wang, Northwestern University — *Evanston, IL*
- Honors thesis (in progress): naysayer proofs for post-quantum signature verification on blockchains
- Selected parameters for and benchmarked a lattice-based commitment construction based on homomorphic Merkle trees

Researcher, Formal Methods & Algebraic Structures — September 2025 – August 2026
Advisor: Prof. Dmitrii Pasechnik, Northwestern University — *Evanston, IL*
- Formalized strongly regular graphs in the Rocq proof assistant, proving core parameter identities and complement closure using the MathComp library
- Implemented certified constructions of Paley and Hamming (rook's) graphs with verified extraction to OCaml, and built a reproducible test pipeline for validating instances

Teaching

Peer Mentor (TA) — Fall 2026
Northwestern University — *Evanston, IL*
- CS 212: Mathematical Foundations of Computer Science (Discrete Math)
- CS 180: Software Design & Composition, assisting with course development

Talks & Presentations

Character Theory of Finite Groups — Spring 2026
- Directed Reading Program capstone talk concluding a year-long sequence. Covered material from Isaacs' *Character Theory of Finite Groups*, including Clifford theory, the Frobenius–Schur indicator, and induction formulas

Representation Theory of Finite Groups — Winter 2026
- Directed Reading Program talk on representation-theory foundations (via *A First Course in Representation Theory*), which led into the character-theory talk above

Formalization of Strongly Regular Graphs in Rocq — Fall 2025
- Presented at Northwestern undergraduate CS research track expo, demonstrating certified constructions and verified extraction pipeline

Selected Coursework

Cryptography & Security: Introduction to Cryptography, Advanced Topics in Modern Cryptography (graduate), Blockchains & Decentralization
Computer Science: Graduate Algorithms (in progress), Design & Analysis of Algorithms, Theory of Computation, Programming Languages, Programming Massively Parallel Processors, Machine Learning
Mathematics: Abstract Algebra (3 quarters), Complex Analysis, Real Analysis (in progress), Linear Algebra: Second Course
Directed Reading: Representation Theory (Fall 2025/Winter 2026), Character Theory (Spring 2026), Elliptic Curves (Summer 2026)
Logic: Elementary Logic II

Industry Experience

Software Engineering Intern June 2026 – August 2026
AlphaSense *Chicago, IL*

- Built features on the Client & Expert Experience team across two large production repositories in React, Vue, and Ruby behind a GraphQL middle layer
- Implemented end-to-end changes spanning GraphQL schema and resolvers through to the client

Software Engineering Intern June 2025 – April 2026
W.R. Berkley – Intrepid Direct Insurance *Overland Park, KS*

- Owned a Salesforce Batch Apex job automating the rollover of renewal coverage terms, eliminating per-policy manual entry
- Built round-robin account assignment logic and internal tooling in .NET Core, Angular, and Apex

Software Engineering Intern May 2024 – August 2024
WellSky *Overland Park, KS*

- Shipped UI and backend improvements to a clinical workflow platform serving 2000+ healthcare providers
- Implemented a patient management feature in Ext JS for pharmacy clients, enabling them to track patient medication adherence

Projects

Galois Playground | *Python, React, SageMath, FastAPI*

- Web application using SageMath (via a FastAPI backend) to visualize group structure, subgroup lattices, and quotient groups

Leadership

President, Blockchain Group at Northwestern April 2026 – Present
Northwestern University *Evanston, IL*

- Revived the dormant group by re-registering it, writing its constitution, securing SAFC funding, and recruiting three officers
- Lead a member project implementing naysayer proofs on Ethereum, from batched Merkle claims to optimistic FRI verification
- Oversee a member-built on-chain strategy game with commit-reveal rounds and verifiable randomness

Technical Skills

Languages: Python, C/C++, Rust, C#, Solidity, Ruby, Java, TypeScript, SQL, Apex
Mathematical Software: SageMath, GAP, Rocq/Coq, LaTeX
Tools & Frameworks: Git, Linux, CUDA, React, Vue, Angular, GraphQL, .NET, Node.js, Rails

Personal Interests

Rock climbing, recorder, basketball, game development